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# Subhalingam D

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## Education

- **Indian Institute of Technology, Delhi** **Jul 2018 – May 2022**
  Bachelor of Technology in Mathematics and Computing | *CGPA: 8.196/10*
- **Chennai Public School, Chennai** **Jun 2016 – May 2018**
  All India Senior School Certificate Examination (CBSE Std. XII) | *Percentage: 96.4%*
- **Chennai Public School, Chennai** **Jun 2014 – May 2016**
  All India Secondary School Examination (CBSE Std. X) | *CGPA: 10/10*

## Experience

**KnowDis AI, Delhi** **May 2022 – Present**
Data Scientist

**Product Category Prediction for Automated Product Mapping (for a Large B2B Marketplace)**

- Designed a retriever-reranker pipeline to classify products into one of **100K+ categories** using metadata like title and specifications
- Curated training data from a **120M+ product catalog** and augmented it with task-specific heuristics to improve data quality
- Fine-tuned a decoder-based retriever; aggregated results using Mean Reciprocal Rank to shortlist candidates across multiple retrievers
- Developed few-shot prompts for **Qwen LLM reranker** for **single-step decoding**; optimized latency using **prefix-caching** and **vLLM**
- Achieved **13% accuracy gain**, **reduced grossly wrong predictions by 66%**, and maintained 92%+ high-confidence coverage
- Integrated the end-to-end system; led the dockerization and coordinated client delivery & testing; solution **deployed to production**

**Attribute-Value Extraction from Product Titles & User Queries (for a Large B2B Marketplace)** *(Research accepted at KDD 2025)*

- Generated weakly supervised training data with **incomplete labeling** from product specifications covering 25K+ attributes
- Designed a novel two-stage system that employs a **marker-augmented generative model** to identify potential attributes, followed by a **cross-encoder-based token classification model** that determines the associated values for each attribute
- **Regenerated training data** to expand attribute-value annotations and **trained a faster NER classifier** on the enriched dataset
- Improved **recall by 20%** while maintaining 90% precision; used for **dynamic feature highlighting** to enhance search experience

**Additional Projects in E-commerce Domain**

- Enhanced the **spell correction model** by augmenting input with lexically similar product titles to better handle low-frequency terms
- **Dockerized a product search system** with large disk-based ANN indexes, deployed using NVIDIA Triton Inference Server
- Developed a **shopping assistant** to enhance user experience via conversational product recommendations and discovery
- Building an agentic system to automatically audit and approve products when sellers list them on the platform

**Style-Controllable English-to-Hindi Translator**

- Extended an encoder-decoder model with **style tokens** to control translation style; fine-tuned on in-house parallel corpora
- Obtained English translations for scraped style-specific monolingual data using **Google Translate API** to augment the training data
- Developed a **style classifier** to sample representative training examples, leading to **improved in-style words usage** in translations

**Projects in Drug Discovery Domain**

- Integrated gradient guidance into de novo molecular generation to better optimize drug-relevant properties *(ICML 2024 Workshop)*
- Developed a "bad rings" detector to filter out generated molecules with non-drug-like ring structures

## Skills

| | |
|---|---|
| **Languages:** | **Python**, Java, C++, C, Bash, MATLAB |
| **ML/AI Frameworks:** | **PyTorch**, Transformers, LangChain, LangGraph, PyTorch-Lightning, TRL, TensorFlow, Keras, NLTK, spaCy |
| **Development:** | **FastAPI**, **AWS**, **NVIDIA Triton**, **vLLM**, ONNX, Streamlit, SQL, Docker, Git, Django |

## Publications

1. **Subhalingam**, **D.**, Kolluru, K., Mausam & Singal, S. *A Framework for Leveraging Partially-Labeled Data for Product Attribute-Value Identification* in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1* (Association for Computing Machinery, 2025). https://doi.org/10.1145/3690624.3709427.
2. Dorna, V., **Subhalingam**, **D.**, Kolluru, K., Tuli, S., Singh, M., Singal, S., Krishnan, N. M. A. & Ranu, S. *TAGMol: Target-Aware Gradient-guided Molecule Generation* in *ICML'24 Workshop ML for Life and Material Science* (2024). https://openreview.net/forum?id=tTtHBSGikn.
3. Goyal, S., Pandey, P., Gaur, G., **D**, **S.**, Bedathur, S. & Ramanath, M. *Tracking entities in technical procedures – a new dataset and baselines* 2021. arXiv: 2104.07378 [cs.CL]. https://arxiv.org/abs/2104.07378.

## Internships

**KnowDis AI, Delhi** **Jan 2022 – May 2022**

Data Science Intern | *Product Category Prediction for Product Search (for IndiaMART)*

- Built a transformer-based **classifier** to predict the most relevant product category from **100K+ labels** using bootstrapped search logs
- Designed heuristics to **enhance atomic label representations** and sampled data to improve category distribution and coverage
- Achieved similar accuracy (**~88%**) as the previous seq2seq model while significantly **reducing average response time** (**3x faster**) and completely **eliminating timeouts**; the model was **integrated with *IndiaMART*'s search system** and was deployed in production

**Samsung R&D Institute, Delhi** **Jun 2021 – Jul 2021**

Software Engineering Intern | *Acoustic Sound Source Localization, Tracking and Separation*

- Developed **sound source direction estimation** module using time delay of arrival of signals between pairs of microphones in an array
- Added modules for tracking active sound sources and extracting individual signals for downstream object identification pipeline
- Integrated stationary noise estimation module for ambient noise removal and reduced maximum direction of arrival error to 7°
- *Received **Pre-Placement Offer (PPO)** for impeccable performance during the internship*

**MateRate Education, Delhi** **May 2020 – Jul 2020**

Web Development Intern | *Results Portal Development and Deployment*

- Designed database schema and built Web APIs using **Django REST framework** to display students' performance reports to parents
- Deployed *Django* backend using **Elastic Beanstalk** with *MySQL* on **RDS** and *React* frontend to **S3** with **CloudFront** CDN integration
- Set up **Auto Scaling group** and attached **Load Balancer** for horizontal scaling; the portal went live with the results of 5000+ students

## Projects

**Tracking State Changes for Entities in Technical Procedural Text** **Feb 2021 – Apr 2022**

*Prof. Srikanta Bedathur and Prof. Maya Ramanath, Research Project (under IBM AI Horizons Network)* *[Paper]*

- Prepared a dataset consisting of ***How-to*** troubleshooting FAQs by **scraping WikiHow pages** from *Computers and Electronics* category
- Constructed baselines using **BERT** & **WordNet** to **predict changes in properties of the entities** involved at each step of the process
- Surveyed the literature to build **next-step recommender** from a given sequence of performed actions and developed LSTM baselines

**Identification of Hate Spreaders on Social Media** **Jan 2022 – Apr 2022**

*Prof. Niladri Chatterjee, Bachelor's Thesis* *[Thesis]*

- Identified key features to profile hate spreaders on Twitter using their feeds and observed high feature importance for sentiment scores
- Proposed a novel scheme that uses **GloVe** embeddings for encoding and **sentiment scores** as weights to quantify word importance
- Attained an **accuracy of 76%** (for English language) on the *PAN@CLEF 2021* dataset *(+1% than best)* and 77% with an ensemble

**Rule-based Written-to-Spoken Text Converter** **Aug 2021 – Sep 2021**

*Prof. Mausam, Natural Language Processing Course*

- Developed a system to rewrite sentences in the spoken form, accounting for dates, times, abbreviations, numerical quantities & units
- Refined the rules through iterative error analysis to handle edge cases (like inflections) and achieved **97.94% F1-score (top in class)**

**Corporate Bankruptcy Prediction** **Feb 2021 – Apr 2021**

*Prof. Niladri Chatterjee, Data Mining Course* *[Report]*

- Inspected bankruptcy prediction models and observed poor recall; hypothesized **class imbalance** & **missing values** to be the reasons
- Trained an ensemble model with **Mean Imputation** & **SMOTE** transformations on *Polish companies* dataset and **gained +10% recall**

**Extended Vector Space Model for News Articles Retrieval** **Oct 2020 – Nov 2020**

*Prof. Srikanta Bedathur, Information Retrieval Course*

- Created an end-to-end retrieval system indexed using **TF-IDF weights** with support for prefix search & named-entity based filters
- Reduced index size by half with **gap encoding**; applied pseudo-relevance feedback based probabilistic **query expansion** for reranking

## Relevant Coursework

Natural Language Processing, Information Retrieval and Web Search, Data Mining, Linguistics (*via* Intro to Language Sciences; Language and Communication), Data Structures and Algorithms, Probability and Stochastic Processes, Statistical Methods, Linear Algebra, Calculus, Fuzzy Sets and Applications, Operating Systems, Differential Equations, Optimization Methods, Theory of Computation

## Activities

- Invited as **Reviewer** for KDD 2026 Datasets and Benchmarks Track (August) — Aug 2025 – Present
- Invited as **Emergency Reviewer** for ACL Rolling Review - February 2025 — Mar 2025 – Apr 2025
- **General Secretary,** Mathematics Society, IIT Delhi — Aug 2021 – Jun 2022
- **Teaching Assistant,** Information Retrieval and Web Search *(Graduate course, Prof. Srikanta Bedathur)* — Aug 2021 – Dec 2021
- **Web Developer,** Student Activity Council, IIT Delhi — Feb 2020 – Sep 2020
- **Volunteer in Teaching projects,** National Service Scheme (NSS), IIT Delhi — Sep 2018 – May 2020